



Gemini 2.5 Flash & 2.5 Flash Image Model Card

Gemini 2.5 Flash - Model Card

Model Cards are intended to provide essential information on Gemini models, including known limitations, mitigation approaches, and safety performance. Model cards may be updated from time-to-time; for example, to include updated evaluations as the model is improved or revised.

Technical Reports are similar to academic papers, and describe models' capabilities, limitations and performance benchmarks. The [Gemini 2.5 technical report](#) contains additional details about the Gemini 2.5 series of models. We recommend that readers seeking more details and information about these models navigate to the technical report.

Last updated: September 26, 2025

Model Information

Description: Gemini 2.5 Flash is the next iteration in the Gemini 2.0 series of models, a suite of highly-capable, natively multimodal, reasoning models. Gemini 2.5 Flash is Google's first fully hybrid reasoning model, giving developers the ability to turn a model's [thinking](#) on or off. The model also allows developers to set thinking budgets to find the right tradeoff between quality, cost, and latency. This model card has been updated to include image as an additional output of Gemini 2.5 Flash; datapoints, evaluations, and information specific to this new capability are specified as 'Gemini 2.5 Flash Image' which has image generation and editing capabilities. Deployment status continues to be "general availability."¹

As of September 25, 2025, there is a preview version offered alongside Gemini 2.5 Flash, specified as "Gemini 2.5 Flash Preview (09-2025)", is designed to deliver a boost in quality, addressing direct developer feedback.

Inputs: Text strings (e.g., a question, a prompt, document(s) to be summarized), images, audio, and video files, with a 1M token context window.

Outputs:

- Gemini 2.5 Flash: Text, with a 64K token output
- Gemini 2.5 Flash Image: Image, with a 32k token output

Architecture: The Gemini 2.5 models are sparse mixture-of-experts (MoE) ([Clark et al., 2022](#); [Du et al., 2021](#); [Fedus et al., 2021](#); [Jiang et al., 2024](#), [Lepikhin et al., 2020](#); [Riquelme et al., 2021](#); [Roller et al., 2021](#); [Shazeer et al., 2017](#); transformers ([Vaswani et al., 2017](#)) with native multimodal support for

¹ We've updated the naming convention throughout this model card to reflect that Gemini 2.5 Flash Image is generally available.

text, vision, and audio inputs. Sparse MoE models activate a subset of model parameters per input token by learning to dynamically route tokens to a subset of parameters (experts); this allows them to decouple total model capacity from computation and serving cost per token. Developments to the model architecture contribute to the significantly improved performance of Gemini 2.5 compared to Gemini 1.5 Pro (see [Section 3](#) of the Gemini Technical Report).

Model Data

Training Dataset: The pre-training dataset was a large-scale, diverse collection of data encompassing a wide range of domains and modalities, which included publicly-available web-documents, code (various programming languages), images, audio (including speech and other audio types) and video. The post-training dataset consisted of vetted instruction tuning data and was a collection of multimodal data with paired instructions and responses in addition to human preference and tool-use data.

Training Data Processing: Data filtering and preprocessing included techniques such as deduplication, safety filtering in-line with [Google's commitment to advancing AI safely and responsibly](#) and quality filtering to mitigate risks and improve training data reliability.

Implementation and Sustainability

Hardware: Gemini 2.5 Flash and Gemini 2.5 Flash Image were trained using [Google's Tensor Processing Units](#) (TPUs). TPUs are specifically designed to handle the massive computations involved in training LLMs and can speed up training considerably compared to CPUs. TPUs often come with large amounts of high-bandwidth memory, allowing for the handling of large models and batch sizes during training, which can lead to better model quality. TPU Pods (large clusters of TPUs) also provide a scalable solution for handling the growing complexity of large foundation models. Training can be distributed across multiple TPU devices for faster and more efficient processing.

The efficiencies gained through the use of TPUs are aligned with Google's [commitment to operate sustainably](#).

Software: Training was done using [JAX](#) and [ML Pathways](#).

Evaluation

Note: all Eval results are only for Gemini 2.5 Flash, unless specifically stated they are for Gemini 2.5 Flash Image.

Approach: Gemini 2.5 Flash was evaluated using the methodology below:

- **Gemini results:** All Gemini 2.5 Flash scores are pass @1 (no majority voting or parallel test time compute unless indicated otherwise). They were all run with the AI Studio API for the model-id gemini-2.5-flash-preview-09-2025, gemini-2.5-flash-preview-05-20, model-id gemini-2.5-flash-preview-04-17 and gemini-2.0-flash with default sampling settings. To reduce variance, we averaged over multiple trials for smaller benchmarks. Vibe-Eval results were reported using Gemini as a judge.
- **Non-Gemini results:** All the results for non-Gemini models were sourced from providers' self-reported numbers unless mentioned otherwise below. All SWE-bench Verified numbers follow official provider reports, using different scaffoldings and infrastructure. Google's scaffolding includes drawing multiple trajectories and re-scoring them using the model's own judgement.
- **Thinking vs not-thinking:** For Claude 3.7 Sonnet: GPQA, AIME 2024, MMMU came with 64k extended thinking, Aider with 32k, and HLE with 16k. Remaining results came from the non thinking model due to result availability. For Grok-3, all results came with extended reasoning except for SimpleQA (based on xAI reports) and Aider.
- **Single attempt vs multiple attempts:** When two numbers were reported for the same evaluation, the higher number used majority voting with n=64 for Grok models and internal scoring with parallel test time compute for Anthropic models.
- **Results sources:** Where provider numbers were not available we reported numbers from leaderboards reporting results on these benchmarks: Humanity's Last Exam results were sourced from [here](#) and [here](#), [AIME 2025 numbers](#), [LiveCodeBench results](#) (10/1/2024 - 2/1/2025 in the UI), [Aider Polyglot numbers](#) and [FACTS](#). For [MRCR](#) v2 we included 128k results as a cumulative score to ensure they can be comparable with previous results and a pointwise value for 1M context window to show the capability of the model at full length. For 2.5 Flash Preview (09-2025) the methodology has changed in this table vs previously published results for MRCR v2 as we have decided to focus on a harder, 8-needle version of the benchmark going forward. We also start measuring SimpleQA Verified instead of SimpleQA for a higher eval signal. See <https://www.kaggle.com/benchmarks/deepmind/simpleqa-verified> for more details.

Results: Gemini 2.5 Flash and 2.5 Flash Preview (09-2025) demonstrated strong performance across a range of benchmarks. Detailed results as of May and September, 2025 are listed below. Additional information on Gemini 2.5 Flash Image follows this table.

Capability Benchmark ²		Gemini 2.5 Flash Preview (09-2025) Thinking	Gemini 2.5 Flash GA Thinking	Gemini 2.5 Flash Preview (04-17) Thinking	Gemini 2.0 Flash Non-thinki ng	OpenAI o4-mini	Claude 3.7 Sonnet 64k Extended thinking	Grok 3 Beta Extended thinking	DeepSe ek R1
Reasoning & knowledge Humanity's Last Exam (no tools)		13.2%	11.0%	12.1%	5.1%	14.3%	8.9%	—	8.6%*
Science GPQA diamond	single attempt (pass@1)	80.8%	82.8%	78.3%	60.1%	81.4%	78.2%	80.2%	71.5%
	multiple attempts	—	—	—	—	—	84.8%	84.6%	—
Mathematics AIME 2025	single attempt (pass@1)	75.6%	72.0%	78.0%	27.5%	92.7%	49.5%	77.3%	70.0%
	multiple attempts	—	—	—	—	—	—	93.3%	—
Code generation LiveCodeBench v5	single attempt (pass@1)	71.7%	63.9%	63.5%	34.5%	—	—	70.6%	64.3%
	multiple attempts	—	—	—	—	—	—	79.4%	—
Code editing Aider Polyglot		—	61.9% / 56.7% whole / diff	51.1% / 44.2% whole / diff	22.2% whole	68.9% / 58.2% whole / diff	64.9% diff	53.3% diff	56.9% diff
Agentic Coding SWE-Bench Verified		54.0%	60.4%	—	—	68.1%	70.3%	—	49.2%
Factuality SimpleQA		25.3%	26.9%	29.7%	29.9%	—	—	43.6%	30.1%
Factuality SimpleQA Verified		27.8%	—	—	—	—	—	—	—
Factuality		87.5%	85.3%	—	84.6%	62.1%	78.8%	74.8%	56.8%

²We regularly update evaluation processes to include new and emerging quality evaluations and benchmarks. The results reported above include additional or updated benchmarks which may not have been included in previous Gemini model cards. Results are thus not directly comparable with performance results found in previous Gemini model cards.

FACTS Grounding									
Visual reasoning MMMU	single attempt (pass@1)	80.3%	79.7%	76.7%	71.7%	81.6%	75.0%	76.0%	no MM support
	multiple attempts	—	—	—	—	—	—	78.0%	no MM support
Image understanding Vibe-Eval (Reka)		62.0%	65.4%	62.0%	56.4%	—	—	—	no MM support
Long context MRCR v2	128k (average)	—	74%	—	36%	49%	—	54%	45%
	1M (pointwise)	—	32%	—	6%	—	—	—	—
Long context MRCR v2 (8-needle)*	128k (average)	52.4%	—	—	—	—	—	—	—
	1M (pointwise)	16.3%	—	—	—	—	—	—	—
Multilingual performance Global MMLU (Lite)		87.9%	88.4%	88.4%	83.4%	—	—	—	—

* indicates evaluated on text problems only (without images)

Approach: Gemini 2.5 Flash Image was evaluated using the methodology below:

- **Human evaluations** of several different quality aspects of text-to-image generation were conducted, including overall preference, prompt-image alignment and visual appeal. Automatic evaluation metrics were used to measure prompt-image alignment and image quality.
- **Result Sources:** human evaluations were conducted via [GenAI-Bench](#) and [LMArena](#)³, including leaderboards reporting provider results on these benchmarks:
 - **Text-to-Image:** Overall Preference, Visual Quality, Text-to-Image Alignment
 - **Image Editing:** Character, Creative, Infographics, Object / Environment, Product Recontextualization, and Stylization

Results: Gemini 2.5 Flash Image demonstrated strong performance across a range of benchmarks. Detailed results as of August 2025 are listed below.

³ While the results reported above from LMArena are not exhaustive of all evaluated models, Gemini 2.5 Flash Image ranked #1 for Text-to-Image and Image Editing as of August 25, 2025.

Text-to-Image:

Capability Benchmark	Gemini Flash 2.5 Image	Imagen 4 Ultra 06-06	ChatGPT 4o / GPT Image 1 (High)	FLUX.1 Kontext [max]	Gemini Flash 2.0 Image
Overall Preference (LMArena)	1147	1135	1129	1075	988
Visual Quality (GenAI-Bench)	1103	1094	1013	864	926
Text-to-Image Alignment (GenAI-Bench)	1042	1053	1046	937	922

Image Editing:

Capability Benchmark	Gemini Flash 2.5 Image	ChatGPT 4o / GPT Image 1 (High)	FLUX.1 Kontext [max]	Qwen Image Edit	Gemini Flash 2.0 Image
Overall Preference (LMArena)	1362	1170	1191	1145	1093
Character	1170	1059	1010	911	850
Creative	1112	1057	968	983	879
Infographics	1067	1029	967	1012	925
Object / Environment	1064	1023	1002	1010	901
Product Recontextualization	1128	1032	943	1009	888
Stylization	1062	1165	949	1091	733

Intended Usage and Limitations

Benefit and Intended Usage:

Gemini 2.5 Flash is well suited for applications that require:

- cost-efficient thinking;
- well-rounded capabilities.

Gemini 2.5 Flash Preview (09-2025) is updated to improve:

- agentic tool use
- improved cost-efficient thinking from Gemini 2.5 Flash

Gemini 2.5 Flash Image augments Gemini 2.5 Flash's benefits and use cases, providing image generation with maintained character and style consistency, prompt-based image editing, native world knowledge, editing of high-quality, high-resolution images in a wide range of visual styles, and multi-image fusion.

Known Limitations: Gemini 2.5 Flash may exhibit some of the general limitations of foundation models, such as hallucinations, and limitations around causal understanding, complex logical deduction, and counterfactual reasoning. Adherence to thinking budgets may not be consistent. Additionally, Gemini 2.5 Flash Image may exhibit some general limitations of text-to-image models such as long-form text rendering and factual representation of fine details in images. The knowledge cutoff date for Gemini 2.5 Flash and Gemini 2.5 Flash Image was January 2025. See the Ethics and Safety section below for additional information on known limitations.

Ethics and Safety

Evaluation Approach: Gemini 2.5 Flash and Gemini 2.5 Flash Image were developed in partnership with internal safety, security, and responsibility teams. A range of evaluations and red teaming activities were conducted to help improve the model and inform decision-making. These evaluations and activities align with [Google's AI Principles](#) and [responsible AI approach](#).

Evaluation types included, but were not limited to:

- **Training/Development Evaluations** including automated and human evaluations carried out continuously throughout and after the model's training, to monitor its progress and performance
- **Human Red Teaming** conducted by specialist teams across the policies and desiderata, deliberately trying to spot weaknesses and ensure the model adheres to safety policies and desired outcomes

- **Automated Red Teaming** to dynamically evaluate Gemini for safety and security considerations at scale, complementing human red teaming and static evaluations
- **Assurance Evaluations** conducted by human evaluators independent of the model development team, and assess responsibility and safety governance decisions
- **Ethics & Safety Reviews** were conducted ahead of the model's release

In addition, we performed testing following the guidelines in [Google DeepMind's Frontier Safety Framework](#) (FSF).

Safety Policies: Gemini safety policies align with Google's standard framework for the types of harmful content that we make best efforts to prevent our Generative AI models from generating, including the following types of harmful content:

1. Child sexual abuse and exploitation
2. Hate speech (e.g., dehumanizing members of protected groups)
3. Dangerous content (e.g., promoting suicide, or instructing in activities that could cause real-world harm)
4. Harassment (e.g., encouraging violence against people)
5. Sexually explicit content
6. Medical advice that runs contrary to scientific or medical consensus

Training and Development Evaluation Results: Results for some of the internal safety evaluations conducted during the development phase are listed below. The evaluation results are for automated evaluations and not human evaluation or red teaming, and scores are provided as an absolute percentage increase or decrease in performance in comparison to the indicated model, as described below.

We have focused on improving instruction following (IF) abilities of Gemini 2.5. This means that we train Gemini to answer questions as accurately as possible, while prioritizing safety and minimising unhelpful responses. New models are more willing to engage with prompts that previous models may have incorrectly refused.

We expect variation in our automated safety evaluations results, which is why we review flagged content to check for egregious or dangerous material. Our manual review confirmed losses were overwhelmingly either a) false positives or b) not egregious and narrowly concentrated around **explicit** requests to produce sexually suggestive content or hateful content, mostly in the context of creative use-cases (e.g. historical fiction).

We continue to improve our internal evaluations, including refining automated evaluations to reduce false positives and negatives, as well as update query sets to ensure balance and maintain a high standard of results. The performance results reported below are computed with improved evaluations and thus are not directly comparable with performance results found in previous Gemini model cards. In addition to continuing to improve our evaluations, we also leverage expert red teamers to assess the safety profile of our models (see below section).

For safety evaluations, a decrease in percentage represents a reduction in violation rates compared to Gemini 2.0 Flash and an increase in percentage represents an increase in violation rates. For tone and instruction following, a positive percentage increase represents an improvement in the tone of the model on sensitive topics and the model's ability to follow instructions while remaining safe compared to Gemini 2.0 Flash. We mark improvements in green and regressions in red.

Evaluation ⁴	Description	Gemini 2.5 Flash Preview Non-thinking (09-2025) vs. Gemini 2.0 Flash	Gemini 2.5 Flash Preview Thinking (09-2025) vs. Gemini 2.0 Flash	Gemini 2.5 Flash Non-thinking (06-17) vs. Gemini 2.0 Flash	Gemini 2.5 Flash Thinking (06-17) vs. Gemini 2.0 Flash
Text to Text Safety	Automated content safety evaluation measuring safety policies	+9.1% (non egregious)	+9.1% (non egregious)	+4.2% (non egregious)	+3.8% (non egregious)
Multilingual Safety	Automated safety policy evaluation across multiple languages	+12.0% (non egregious)	+12.0% (non egregious)	+13.1% (non egregious)	+11.9% (non egregious)
Image to Text Safety	Automated content safety evaluation measuring safety policies	+6.0% (non egregious)	+4.8% (non egregious)	+3.7% (non egregious)	+1.6% (non egregious)
Tone	Automated evaluation measuring objective tone of model refusal	+9.5%	+9.7%	+0.7%	-2.3%
Instruction Following	Automated evaluation measuring model's ability to follow instructions while remaining safe	+29.2%	+30.4%	+28.1%	+28.4%

Assurance Evaluations Results: We conduct baseline assurance evaluations to guide decisions on model releases. These evaluations look at model behavior, including within the context of the safety policies and modality-specific risk areas. High-level findings are fed back to the model team. For content safety policies, including child safety, we saw similar or improved safety performance compared to Gemini 2.0 Flash.

Frontier Safety Assessment: We evaluated Gemini 2.5 Pro Preview for Frontier Safety and reported the results in the [2.5 Pro Preview model card](#), finding that it did not reach any critical capability levels outlined in our [Frontier Safety Framework](#). As Gemini 2.5 Flash is less capable than Gemini 2.5 Pro Preview, and the Gemini 2.5 Pro model results give us confidence that Gemini 2.5

⁴The ordering of evaluations in this table has changed from previous iterations of the 2.5 Flash-Lite model card in order to list safety evaluations together and improve readability. The type of evaluations listed have remained the same.

Flash is unlikely to reach critical capability levels, we can rely on Frontier Safety evaluations reported for Gemini 2.5 Pro Preview.

Known Safety Limitations: The main safety limitations for Gemini 2.5 Flash are related to tone. The model will sometimes respond in a way which can come across as “preachy”. However, Gemini 2.5 Flash still has measurable improvements in tone over previous Flash models.

Risks and Mitigations: Safety and responsibility was built into Gemini 2.5 Flash throughout the training and deployment lifecycle, including pre-training, post-training, and product-level mitigations. Mitigations include, but are not limited to:

- dataset filtering
 - conditional pre-training
 - supervised fine-tuning
 - reinforcement learning from human and critic feedback
 - safety policies and desiderata
 - product-level mitigations such as safety filtering
-